Example 6.13

Find the value of $\tan 70°13'$

Solution

	0′	6′	12′	18′	24′	30′	36′	42′	48′	54′	Mean Difference				
	0.0°	0.1°	0.2°	0.3°	0.4°	0.5°	0.6°	0.7°	0.8°	0.9°	1	2	3	4	5
70°			2.7776								26				

Write $70°13' = 70°12' + 1'$

From the table we have, $\tan 70°12' = 2.7776$

Mean difference for $1' = 26$ (Mean difference to be added for tan)

$\tan 70°13' = 2.7802$

Example 6.14

Find the value of

(i) $\sin 38°36' + \tan 12°12'$ (ii) $\tan 60°25' - \cos 49°20'$

Solution

(i) $\sin 38°36' + \tan 12°12'$

$\sin 38°36' = 0.6239$

$\tan 12°12' = 0.2162$

$\sin 38°36' + \tan 12°12' = 0.8401$

(ii) $\tan 60°25' - \cos 49°20'$

$\tan 60°25' = 1.7603 + 0.0012 = 1.7615$

$\cos 49°20' = 0.6521 - 0.0004 = 0.6517$

$\tan 60°25' - \cos 49°20' = 1.1098$

Example 6.15

Find the value of θ if

(i) $\sin\theta = 0.9858$ (ii) $\cos\theta = 0.7656$

Solution

(i) $\sin\theta = 0.9858 = 0.9857 + 0.0001$

From the sine table $0.9857 = 80°18'$

Mean difference $1 = 2'$

$0.9858 = 80°20'$

$\sin\theta = 0.9858 = \sin 80°20'$

$\theta = 80°20'$

(ii) $\cos\theta = 0.7656 = 0.7660 - 0.0004$

From the cosine table

$0.7660 = 40°0'$

Mean difference $4 = 2'$

$0.7656 = 40°2'$

$\cos\theta = 0.7656 = \cos 40°2'$

$\theta = 40°2'$

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